Amazon Aurora

Distributed Cloud-Native Database

A Deep Dive into Architecture, Scalability & Fault Tolerance

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What is Amazon Aurora?

Overview

Amazon Aurora is a fully managed, cloud-native relational database engine built by AWS.

Compatible with both MySQL and PostgreSQL, offering up to 5× MySQL and 3× PostgreSQL performance.

Designed from scratch for the cloud — separates compute and storage.

Automatically replicates data 6 ways across 3 Availability Zones for high durability.

Storage auto-scales from 10 GB up to 128 TB with zero downtime.

Launched

2014

by Amazon Web Services

Engine

MySQL + PostgreSQL

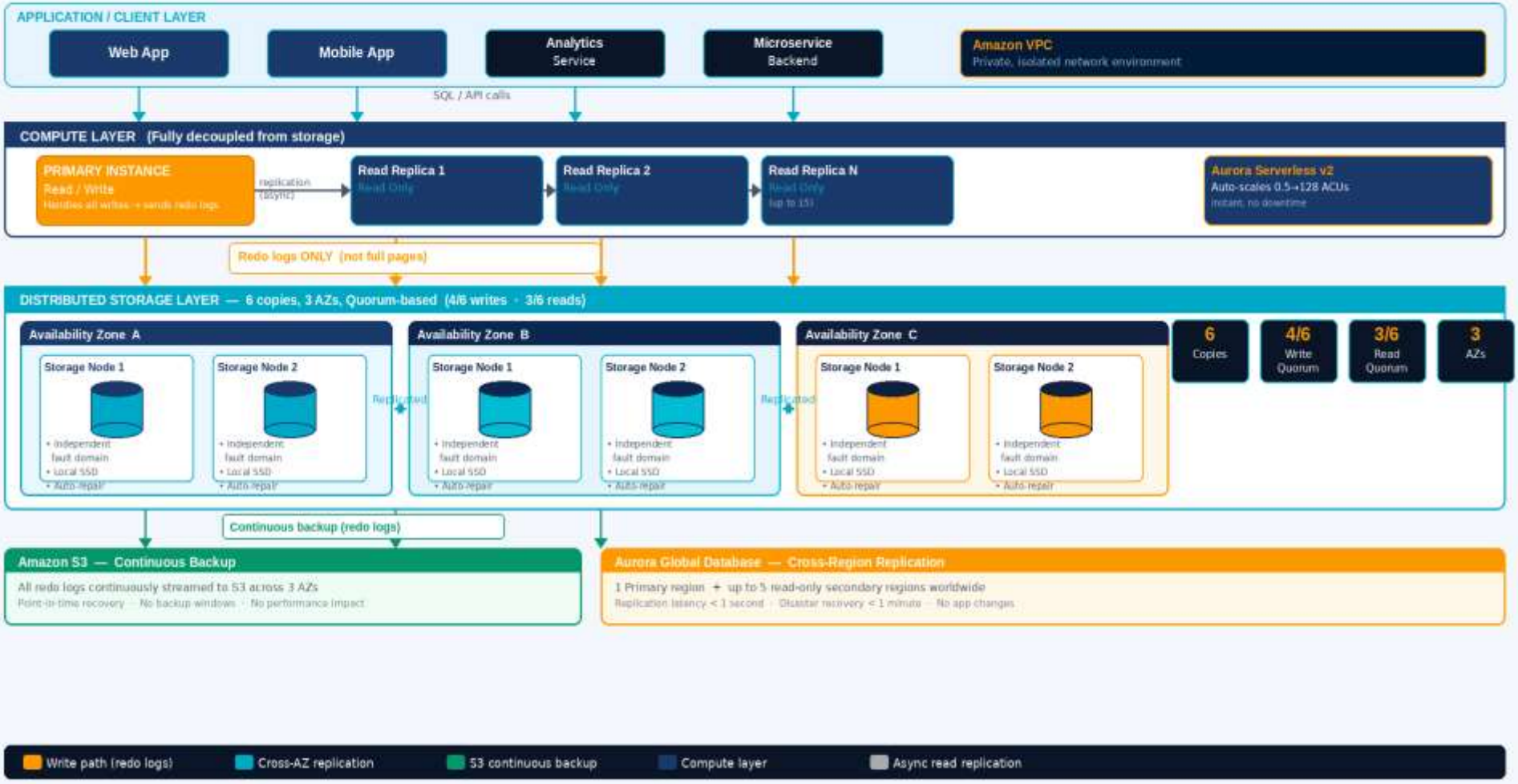
compatible

Storage

Up to 128 TB

auto-scaling

Amazon Aurora — Architecture Overview



Aurora vs Traditional RDS

Feature	Traditional RDS	Amazon Aurora
Replication	Single-region sync	**6 copies / 3 AZs**
Storage Scaling	Manual provisioning	**Auto: 10GB → 128TB**
Failover Time	60–120 seconds	**< 30 seconds**
Read Replicas	Up to 5	**Up to 15**
Performance	Baseline MySQL/PG	**5× MySQL, 3× PG**
Backup	EBS snapshots	**Continuous to S3**
Global Tables	Not supported	**Aurora Global DB**

Key Features of Amazon Aurora

High Performance

5× MySQL, 3× PostgreSQL speed via log-structured distributed storage.

Fault Tolerance

6 data copies across 3 AZs. Tolerates loss of 2 copies for writes, 3 for reads.

Auto-Healing

Continuously scans for errors and auto-repairs data blocks from peers.

Auto Scaling

Serverless v2: instantly scales compute & storage with zero downtime.

Global Database

Cross-region replication with < 1 sec latency. 5 read regions.

Security

Encryption at rest & in transit. VPC isolation, IAM authentication.

Aurora's Distributed Storage Engine

Log-Structured Storage

Aurora sends only redo logs to storage — not full data pages. This reduces network traffic by up to 7.7×.

Quorum Writes

Writes succeed when 4 of 6 storage nodes acknowledge. Reads require 3 of 6 — ensuring consistency.

No Checkpointing

Aurora never writes a checkpoint page. Storage nodes reconstruct pages on demand from logs.

S3 Continuous Backup

Every transaction is continuously backed up to Amazon S3 across 3 AZs — no backup windows needed.

6

Storage Copies

3

Availability Zones

4/6

Write Quorum

3/6

Read Quorum

Aurora in Distributed Systems Context

CAP Theorem Positioning

Aurora prioritizes Consistency + Partition-Tolerance (CP). During a network partition, writes halt until quorum is restored.

Eventual vs Strong Consistency

Primary instance: strong read consistency. Read replicas: slightly eventual (< 100ms lag). Configurable per-query.

Consensus & Replication

Uses a custom quorum-based consensus. 4-of-6 write quorum ensures data durability even if an entire AZ fails.

Failure Detection & Recovery

Self-healing storage detects bit-rot and peer-repairs blocks. Failover completes in < 30s with no data loss.

Aurora Serverless & Global Database

Aurora Serverless v2

Instantly scales from 0.5 to 128 ACUs (Aurora Capacity Units) in < 1 second.

Pauses automatically when idle — pay only for what you use.

No capacity planning required — Aurora handles it automatically.

Ideal for dev/test environments, variable workloads, and SaaS applications.

Supports both MySQL and PostgreSQL engines.

Aurora Global Database

1 primary region + up to 5 read-only secondary regions worldwide.

Cross-region replication latency < 1 second using dedicated replication infrastructure.

Enables low-latency reads for globally distributed users.

Promotes a secondary region to primary in < 1 minute during disaster recovery.

No application changes needed — transparent failover.

Real-World Use Cases

Netflix	Stores metadata for 260M+ subscribers. Uses Aurora's low-latency reads to power content recommendations globally.
Samsung	Powers backend services for SmartThings IoT platform handling billions of events per day with Aurora's scaling.
Airbnb	Uses Aurora for their core booking database. Leverages read replicas to distribute global traffic during peak seasons.
Dow Jones	Migrated news publishing platform to Aurora for 3× performance improvement while cutting database costs by 40%.

Pros & Cons of Amazon Aurora

✓ Advantages

- ✓ Up to 5× MySQL and 3× PostgreSQL performance
- ✓ Automatic 6-way replication across 3 AZs
- ✓ Zero-downtime auto-scaling storage (128 TB max)
- ✓ Continuous, automatic backup to Amazon S3
- ✓ Sub-30 second failover with no data loss
- ✓ Fully managed — no patching, backups, or tuning

✗ Limitations

- ✗ AWS vendor lock-in — not portable to other clouds
- ✗ Higher cost than standard RDS for small workloads
- ✗ Read replicas can have slight eventual consistency lag
- ✗ Limited to MySQL and PostgreSQL engines only
- ✗ Serverless cold-start latency for paused instances
- ✗ Global DB read regions are read-only (not multi-master)

Conclusion

Amazon Aurora redefines what a relational database can be in a distributed cloud environment. By separating compute and storage, leveraging a custom quorum-based storage engine, and offering seamless global replication, Aurora solves the fundamental challenges of distributed systems — fault tolerance, consistency, and scalability — without burdening the developer.

Distributed Storage: 6 copies across 3 AZs with automatic healing

Performance: 5× MySQL speed via log-structured writes

Scalability: Serverless auto-scaling with no downtime

Reliability: < 30s failover, continuous S3 backup

Thank You!

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